

HDOC Day-15

Write a menu driven program using switch-case to input a positive integer and perform one of the following operations based on the user's choice.

- **Check whether all the digits are even.**

E.g-1 Number: 12345, Output: All digits not even

E.g-2 Number:24680, Output: All the digits are even

- **Check whether all digits are prime.**
- **E.g-1 Number: 2357, Output: All digits are prime.**
- **E.g-2 Number: 2379, Output: All digits are not prime.**

1. Algorithm:

A. Check whether all digits are even

Step 1: Start.

Step 2: Input a positive integer n.

Step 3: Store n in a temporary variable temp.

Step 4: Extract the last digit using:

digit = temp % 10

Step 5: Check whether digit is even using digit % 2 == 0.

If false, set allEven = 0.

If true, continue.

Step 6: Remove the last digit:

temp = temp / 10

Step 7: Repeat Steps 4–6 until temp becomes 0.
Step 8: If allEven == 1, display "All digits are even".
Otherwise, display "All digits are not even".
Step 9: Stop.

B. Check whether all digits are prime

The prime digits are:

2, 3, 5, 7

Step 1: Start.

Step 2: Input a positive integer n.

Step 3: Store n in temp.

Step 4: Extract the last digit:

digit = temp % 10

Step 5: Check whether digit is one of 2, 3, 5, 7.

- If yes, continue.**
- If no, set allPrime = 0.**

Step 6: Remove the last digit:

temp = temp / 10

Step 7: Repeat until temp becomes 0.

Step 8: If allPrime == 1, display "All digits are prime".

Otherwise, display "All digits are not prime".

Step 9: Stop

• All digits are even:

Test Case	Input Number	Expected Output
1	12345	All digits are not even
2	24680	All digits are even
3	2222	All digits are even
4	12346	All digits are not even

5	8080	All digits are even
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• All digits are prime:

Test Case	Input Number	Expected Output
1	2357	All digits are prime
2	2379	All digits are not prime
3	2225	All digits are prime
4	1357	All digits are prime
5	1237	All digits are not prime

• C Program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, choice;
```

```
    int temp, digit;
```

```
    int allEven, allPrime;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &n);
```

```
    printf("\nMENU\n");
```

```
    printf("1. Check whether all digits are even\n");
```

```
    printf("2. Check whether all digits are prime\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch(choice)
```

```
{
```

```
    case 1:
```

```
        temp = n;
```

```
        allEven = 1;
```

```
        while(temp > 0)
```

```
        {
```

```
            digit = temp % 10;
```

```
            if(digit % 2 != 0)
```

```
            {
```

```
                allEven = 0;
```

```
                break;
```

```
            }
```

```
            temp = temp / 10;
```

```
        }
```

```
    if(allEven == 1)
```

```
        printf("All digits are even\n");
```

```
    else
```

```
printf("All digits are not even\n");
```

```
break;
```

case 2:

```
temp = n;
```

```
allPrime = 1;
```

```
while(temp > 0)
```

```
{
```

```
    digit = temp % 10;
```

```
    if(digit != 2 && digit != 3 &&
```

```
        digit != 5 && digit != 7)
```

```
    {
```

```
        allPrime = 0;
```

```
        break;
```

```
    }
```

```
    temp = temp / 10;
```

```
}
```

```
if(allPrime == 1)
```

```
    printf("All digits are prime\n");
```

```
else
```

```
printf("All digits are not prime\n");
```

```
break;
```

```
default:
```

```
printf("Invalid choice\n");
```

```
}
```

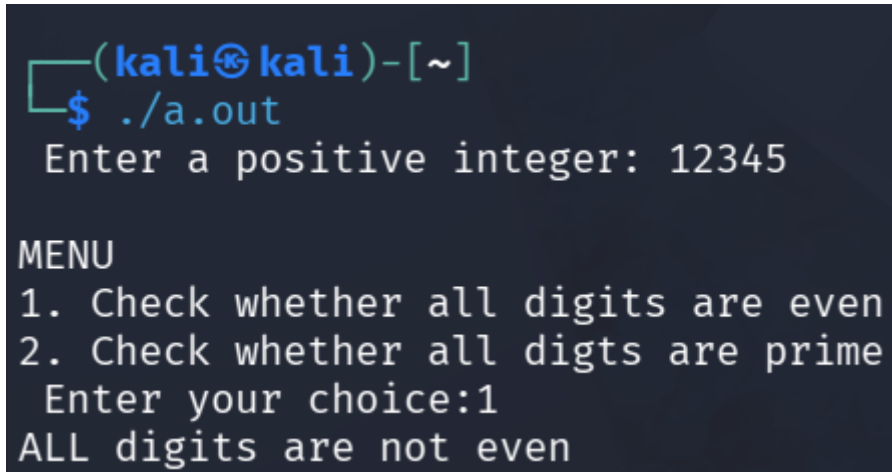
```
return 0;
```

```
}
```

Test Case Table:

Choice 1:

Test Case 1:



```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: 12345  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice:1  
ALL digits are not even
```

Test Case 2:

```
(kali㉿kali)-[~]  
└─$ ./a.out  
Enter a positive integer: 24680  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice:1  
All digits are even
```

Test Case 3:

```
(kali㉿kali)-[~]  
└─$ ./a.out  
Enter a positive integer: 2222  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice:1  
All digits are even
```

Test Case 4:

```
(kali㉿kali)-[~]  
└─$ ./a.out  
Enter a positive integer: 12346  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice:1  
ALL digits are not even
```

Test Case 5:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: 8080  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digts are prime  
Enter your choice:1  
All digits are even
```

For Choice 2:

Test Case 1:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: 2357  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digts are prime  
Enter your choice:2  
All digits are prime
```

Test Case 2:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: 2379  
  
MENU  
1. Check whether all digits are even  
2. Check whether all digts are prime  
Enter your choice:2  
All digits are not prime
```

Test Case 3:

```
(kali@kali)-[~]  
$ ./a.out
```

Enter a positive integer: 2225

MENU

1. Check whether all digits are even
2. Check whether all digits are prime

Enter your choice:2

All digits are prime